

Hybrid CP-SAT and Quantum-Inspired Optimization for Airline Disruption Recovery with Explainable Decision Support

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Abstract

Airline disruption management requires solving tightly coupled combinatorial optimization problems—aircraft assignment, crew scheduling, and delay propagation—under strict regulatory constraints and real-time operational pressure. This paper presents a hybrid decision support framework that integrates Google OR-Tools CP-SAT constraint programming with a Quantum-Inspired Genetic Algorithm (QIGA) through a warm-start mechanism, augmented by a SHAP-based explainability layer for transparent decision-making. The CP-SAT solver generates feasible solutions that strictly enforce EASA Flight Time Limitation (FTL) regulations and operational constraints including aircraft continuity, crew continuity, gate capacity, and minimum connection times. These solutions are subsequently refined by QIGA, which leverages quantum-rotation-gate-driven population evolution to escape local optima within limited computational budgets. An explainability module combining TreeSHAP feature attribution with constraint-driven decision reason codes provides interpretable rationale for each operational decision, addressing EASA AMC 20-42 transparency requirements. Experimental evaluation on a

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150-flight hub-and-spoke network demonstrates that the hybrid approach achieves a 5% improvement in operational profit over standalone CP-SAT ($p = 0.007$), reduces flight cancellations by 20%, and maintains zero FTL violations across all test scenarios. The system converges to solutions within 2.1% of optimality under a 60-second time limit, confirming practical viability for real-time airline operations control.

Keywords: Airline Disruption Management, Constraint Programming, CP-SAT, Quantum-Inspired Genetic Algorithm, Explainable AI, SHAP, EASA Flight Time Limitations, Irregular Operations

1. Introduction

Commercial aviation transported approximately 4.5 billion passengers and executed 8.8 million scheduled flights in 2023 [?]. This operational scale generates tightly coupled decision spaces encompassing flight scheduling, aircraft-crew assignments, airport slot coordination, maintenance windows, and passenger connections—each individually NP-hard in combinatorial complexity [? ?]. Historical NEXTOR estimates put the annual cost of US flight delays at approximately \$32.9 billion (2007 USD), a figure that remains indicative of the systemic scale even after inflation adjustment [?], with significant portions attributable to suboptimal resource allocation and delayed operational decisions rather than external factors [?]. AhmadBeygi et al. [?] demonstrated that crew duty chain continuity plays a critical role in delay propagation across airline networks.

Traditional approaches to disruption management rely on either exact optimization methods—such as Mixed-Integer Linear Programming (MILP) and con-

15 straint programming—or metaheuristic techniques including genetic algorithms
16 and local search. While exact methods provide strong feasibility guarantees and
17 can rigorously enforce operational constraints, they often struggle to scale under
18 the tight time limits imposed by real-world airline operations control centers [? ?
19]. Conversely, metaheuristic approaches offer improved exploration of large solu-
20 tion spaces but may fail to guarantee feasibility with respect to complex regulatory
21 constraints or converge prematurely to suboptimal solutions [? ?].

22 Recent advances in constraint programming, particularly Google OR-Tools
23 CP-SAT, which combines constraint propagation with Boolean satisfiability (SAT)-
24 based conflict-driven clause learning, have demonstrated significant potential in
25 solving large-scale combinatorial scheduling problems [? ?]. In parallel, quantum-
26 inspired evolutionary algorithms have emerged as promising metaheuristics that
27 enhance population diversity through probabilistic representations inspired by quan-
28 tum superposition [? ?].

29 Another critical limitation of existing decision support systems lies in their
30 lack of transparency. The European Union Aviation Safety Agency (EASA) AMC
31 20-42 guideline mandates that AI/ML-based decision systems in aviation provide
32 explainability, traceability, and robustness [?]. SHAP [?], LIME [?], and
33 counterfactual explanations [?] represent post-hoc explainability methods that
34 address these requirements.

35 Individual components of this pipeline (CP-SAT scheduling, quantum-inspired
36 metaheuristics, SHAP attribution) have each appeared in prior work, typically in
37 isolation. The contribution of this paper is not the invention of a new component,

38 but rather a **tightly coupled integration** for which we can offer (i) a formal warm-
 39 start non-degradation guarantee (Proposition ??) that makes the CP-SAT $\rightarrow$ QIGA
 40 hand-off provably safe under a fixed time budget, and (ii) a structured mapping
 41 between optimization constraints and post-hoc explanations aligned with EASA
 42 AMC 20-42 traceability requirements. To our knowledge, no prior study com-
 43 bines all three elements under a single formally-analyzed framework targeting
 44 airline disruption recovery. Unlike classical recovery optimization approaches
 45 that lack explainability [?] or learning-based predictors that cannot enforce hard-
 46 constraint satisfaction [?], the proposed framework ensures **mathematical feasi-**
 47 **bility** of all solutions while providing **auditable decision rationale** aligned with
 48 emerging aviation AI regulation (EASA AMC 20-42). Furthermore, whereas ex-
 49 isting quantum-inspired optimization studies [?] evaluate QIGA in isolation,
 50 we demonstrate that its effectiveness is **contingent on warm-start initialization**
 51 from an exact solver—a finding with implications for hybrid algorithm design
 52 beyond the aviation domain.

53 The main contributions are:

- 54 1. **Hybrid Optimization Framework:** A novel two-stage integration of CP-
 55 SAT and QIGA for airline disruption recovery. We formally prove (Proposi-
 56 tion ??) that warm-start initialization guarantees monotonic non-degradation
 57 of solution quality, combining exact feasibility with enhanced exploratory
 58 search.
- 59 2. **EASA-Compliant Constraint Formulation:** Direct encoding of EASA
 60 CAT.OP.MPA.210 Flight Time Limitations as hard constraints in the CP-

61 SAT model, with automatic generation of decision reason codes via post-
62 solve constraint slack analysis.

63 **3. Integrated Explainable Decision Support:** A dual-layer explainability ar-
64 chitecture combining TreeSHAP feature attribution (predictive transparency)
65 with structured constraint-based decision reasoning (optimization transparency),
66 providing auditable rationale aligned with EASA AMC 20-42.

67 **4. Comprehensive Experimental Evaluation:** Rigorous performance analy-
68 sis with statistical significance testing (Wilcoxon signed-rank, $p = 0.007$),
69 ablation studies, and multi-scenario robustness evaluation demonstrating
70 consistent improvements under diverse disruption conditions.

71 The remainder of this paper is organized as follows. Section ?? reviews re-
72 lated work. Section ?? presents the mathematical formulation. Section ?? de-
73 scribes the hybrid optimization framework. Section ?? details the explainability
74 layer. Section ?? provides experimental results. Section ?? discusses findings and
75 limitations. Section ?? concludes the paper.

76 **2. Related Work**

77 *2.1. Exact Optimization Methods*

78 The classical formulation of crew pairing as a Set Partitioning Problem (SPP)
79 with column generation remains the gold standard for offline planning [? ?].
80 Fleet assignment is typically modeled as MILP [? ? ?]. While these methods
81 guarantee global optimality, solution times range from hours to days for large-
82 scale instances, rendering them impractical for real-time disruption recovery. Yen

83 and Birge [?] extended this framework to stochastic programming, though com-
84 putational costs escalate with scenario count.

85 2.2. *Constraint Programming*

86 Constraint programming offers natural modeling for multi-constrained schedul-
87 ing [?]. Google OR-Tools CP-SAT combines lazy clause generation, conflict-
88 driven clause learning, and portfolio search, achieving top performance in multi-
89 ple MiniZinc Challenge benchmarks [?]. Laborie et al. [?] demonstrated that
90 CP Optimizer’s interval variable abstractions enable more compact modeling for
91 scheduling problems compared to MILP.

92 2.3. *Metaheuristic and Quantum-Inspired Methods*

93 Genetic algorithms for crew scheduling were pioneered by Levine [?] and
94 extended by Ozdemir and Mohan [?]. The Quantum-Inspired Evolutionary Al-
95 gorithm (QIEA) proposed by Han and Kim [?] encodes chromosomes as Q-bits,
96 maintaining population diversity longer than classical GA. Zhang [?] reviewed
97 quantum-inspired evolutionary algorithms across combinatorial domains and re-
98 ported 15–30% faster convergence on large instances relative to classical GAs.

99 2.4. *Explainable AI in Aviation*

100 SHAP [?] provides game-theoretic feature attribution; TreeSHAP [?] com-
101 puts exact Shapley values for tree-based models in polynomial time. Counterfac-
102 tual explanations complement feature attribution by identifying minimal action-
103 able changes that flip a prediction [?].

104 2.5. Hybrid Approaches and Research Gap

105 Rosenberger et al. [?] formulated aircraft rerouting as a large-scale integer
106 program for recovery, but without explainability or learning-based demand pri-
107 ors. Jacquillat and Odoni [?] integrate scheduling and operations for airport
108 congestion mitigation but do not address airline-side crew or FTL constraints.
109 Barredo Arrieta et al. [?] survey XAI taxonomies relevant to safety-critical AI but
110 without integration into combinatorial optimization pipelines. The simultaneous
111 combination of CP-SAT exact optimization with quantum-inspired metaheuristic
112 refinement and integrated multi-level explainability for airline disruption recovery
113 remains unexplored in the literature—a gap this paper addresses.

114 Specifically, three critical gaps emerge from the literature: (i) the **optimality–**
115 **speed–explainability trilemma**, where exact methods guarantee optimality but
116 lack transparency, while learning-based methods are fast but cannot ensure reg-
117 ulatory compliance; (ii) the **absence of open-source EASA FTL integration** in
118 CP models with automatic constraint attribution; and (iii) the **lack of empirical**
119 **evidence** on whether warm-start coupling between exact and quantum-inspired
120 solvers yields synergistic benefits beyond standalone performance.

121 Table ?? provides a comparative positioning of the proposed approach.

Table 1: Comparative positioning of the proposed approach in the literature.

Approach	Source	Problem	Scale	Time	Optimality	XAI
Column Generation	[?]	Crew pairing	≤ 5000	Hours	Global	—
MILP (Gurobi)	[?]	Fleet assign.	≤ 2000	Minutes	Global	—
CP + CBLS	[?]	Scheduling	Medium	Minutes	Near-opt.	—
GA	[?]	Crew pairing	Medium	Seconds	Heuristic	—
QIEA	[?]	Combinatorial	Variable	Seconds	Heuristic	—
ML prediction	[?]	Delay prop.	Network	Offline	Predictive	—
Rerouting IP	[?]	Recovery	Medium	Minutes	Near-opt.	—
This paper	—	IROPS	≤ 500	≤ 60 s	CP+QIGA	✓

122 3. Problem Formulation

123 3.1. Problem Description

124 Airline disruption recovery is modeled as an integrated optimization problem
125 that jointly considers aircraft assignment, crew scheduling, and delay management
126 under operational and regulatory constraints. The system is represented using a
127 time–space network (TSN), where nodes correspond to airport–time pairs and arcs
128 represent feasible transitions such as flight legs, ground waiting, and maintenance
129 activities.

130 3.2. Sets and Decision Variables

131 Let F , K , and C denote the sets of flights, aircraft, and crew, respectively, with
132 A denoting the set of airports and T the set of time slots (15-minute intervals).

Table 2: Decision variables of the CP-SAT model.

Symbol	Type	Description	Domain
$x_{f,k}$	Binary	Flight f assigned to aircraft k	$f \in F, k \in K$
$y_{f,c}$	Binary	Flight f assigned to crew c	$f \in F, c \in C$
z_f	Binary	Flight f is cancelled	$f \in F$
d_f	Integer	Delay assigned to flight f (min)	$0 \leq d_f \leq D_{\max}$
δ_f	Integer	Actual departure time of f	—
$g_{f,a,t}$	Binary	f uses gate at airport a , time t	$(f, a, t) \in \mathcal{G}$

133 3.3. Constraints

134 The model enforces ten constraint families ensuring operational feasibility and
135 regulatory compliance (Table ??).

Table 3: Constraint families.

Constraint	ID	Description	Type
Assignment uniqueness	C ₁	Each flight $\rightarrow$ at most one aircraft	Hard
Crew uniqueness	C ₂	Each flight $\rightarrow$ at most one crew	Hard
Aircraft capacity	C ₃	Capacity $\geq$ passenger count	Hard
Aircraft continuity	C ₄	Same aircraft connects at same airport	Hard
Crew continuity	C ₅	Same crew connects at same airport	Hard
EASA FTL	C ₆	Crew cumulative duty $\leq$ 600 min	Hard
Gate capacity	C ₇	Max 1 aircraft/gate/time slot	Hard
Slot capacity	C ₈	Hourly movements $\leq$ airport capacity	Hard
Min. Connection Time	C ₉	Passenger connection $\geq$ MCT	Hard
Cancel consistency	C ₁₀	Cancelled $\Rightarrow$ no assignments	Hard

136

(C₁) Assignment uniqueness:

$$\sum_{k \in K} x_{f,k} + z_f = 1, \quad \forall f \in F \quad (1)$$

137

(C₂) Crew uniqueness:

$$\sum_{c \in C} y_{f,c} \leq 1 - z_f, \quad \forall f \in F \quad (2)$$

138 **(C₃) Aircraft capacity:**

$$\sum_{k \in K} \text{cap}_k \cdot x_{f,k} \geq \text{pax}_f \cdot (1 - z_f), \quad \forall f \in F \quad (3)$$

139 **(C₄) Aircraft continuity:** For consecutive flights f_1, f_2 on the same aircraft
140 k :

$$x_{f_1,k} + x_{f_2,k} \leq 1 \quad \text{if } \text{dest}_{f_1} \neq \text{orig}_{f_2} \quad (4)$$

141 **(C₆) EASA FTL (conservative proxy):**

$$\sum_{f \in F} y_{f,c} \cdot \text{block}_f \leq \bar{B}, \quad \forall c \in C, \quad (5)$$

142 where $\bar{B} = 600$ min is used as a *conservative uniform ceiling* derived from the
143 tightest single-crew-type entry of EASA CAT.OP.MPA.210 under a one-sector,
144 non-WOCL scenario. This proxy is deliberately restrictive: any schedule satisfy-
145 ing (??) is also feasible under the full FDP $\times$ sector $\times$ WOCL matrix, so the model
146 cannot produce FTL violations when the complete table is substituted. We re-
147 tain (??) in the main experiments because it preserves the combinatorial structure
148 exercised by CP-SAT while yielding a regulator-safe over-approximation; a richer
149 table-driven formulation is a drop-in replacement that we discuss in Section ??.

150 **(C₉) Minimum Connection Time:** For passenger connection $(f_{\text{in}}, f_{\text{out}})$:

$$\delta_{f_{\text{out}}} - (\delta_{f_{\text{in}}} + \text{block}_{f_{\text{in}}}) \geq \text{MCT}_a \cdot (1 - z_{f_{\text{in}}} - z_{f_{\text{out}}}) \quad (6)$$

151 3.4. Objective Function

152 The objective minimizes total operational cost:

$$\min \sum_{f \in F} \left[\alpha \cdot z_f \cdot R_f + \beta \cdot d_f \cdot w_f + \gamma \sum_{k \in K} x_{f,k} \cdot \text{fuel}_{f,k} + \mu \cdot z_f \cdot \text{slotPen}_f \right] \quad (7)$$

153 where R_f is the revenue loss from cancellation, w_f is the delay cost per minute
 154 (weighted by passenger volume and connection criticality), $\text{fuel}_{f,k}$ is the fuel cost,
 155 and $\alpha, \beta, \gamma, \mu$ are strategy-dependent weights (Table ??).

Table 4: Strategy weight configurations.

Strategy	α (cancel)	β (delay)	γ (fuel)	μ (slot)
PROFIT	1.0	1.0	0.3	0.5
ECO	0.8	1.2	1.5	0.4
RESILIENCE	1.5	0.8	0.4	1.0

156 3.5. Complexity Analysis

157 The model has $O(|F| \cdot (|K| + |C|))$ variables and $O(|F|^2)$ constraints. For the
 158 experimental instance ($|F| = 150$, $|K| = 20$, $|C| = 40$): $\sim 9,000$ variables and
 159 $\sim 22,500$ constraints. QIGA complexity: $O(G_{\max} \cdot N \cdot |F|)$ with typical $G_{\max} =$
 160 200 , $N = 50$.

161 4. Hybrid Optimization Framework

162 4.1. Overview

163 The proposed framework operates in two stages (Fig. ??): (i) the CP-SAT
164 solver generates a high-quality feasible solution under strict constraint enforce-
165 ment, and (ii) QIGA refines this solution through population-based evolutionary
166 search initialized via warm-start. When CP-SAT returns `TIMEOUT` or `INFEASIBLE`,
167 the system automatically transitions to QIGA for recovery.

168 4.2. Stage 1: CP-SAT Feasible Solution Generation

169 The first stage employs the CP-SAT solver from Google OR-Tools [?]. CP-
170 SAT combines constraint propagation with SAT-based conflict-driven clause learn-
171 ing, enabling efficient exploration of large combinatorial spaces. The solver op-
172 erates under a configurable time budget (default: 60 seconds) and guarantees that
173 all hard constraints—including aircraft continuity (C_4) and EASA FTL (C_6)—are
174 strictly satisfied. Let S_{CP} denote the best feasible solution obtained within the
175 time limit.

176 4.3. Stage 2: Quantum-Inspired Genetic Algorithm (QIGA)

177 Each individual in the QIGA population is encoded using a vector of Q-bits:

$$q_i = (\alpha_i, \beta_i), \quad |\alpha_i|^2 + |\beta_i|^2 = 1 \quad (8)$$

178 Each chromosome $\xi = (\xi_1, \xi_2, \dots, \xi_n)$ encodes an action vector where $\xi_i \in$
179 $\{\text{KEEP}, \text{DELAY}_{15}, \text{DELAY}_{30}, \text{DELAY}_{60}, \text{CANCEL}, \text{SWAP}\}$.

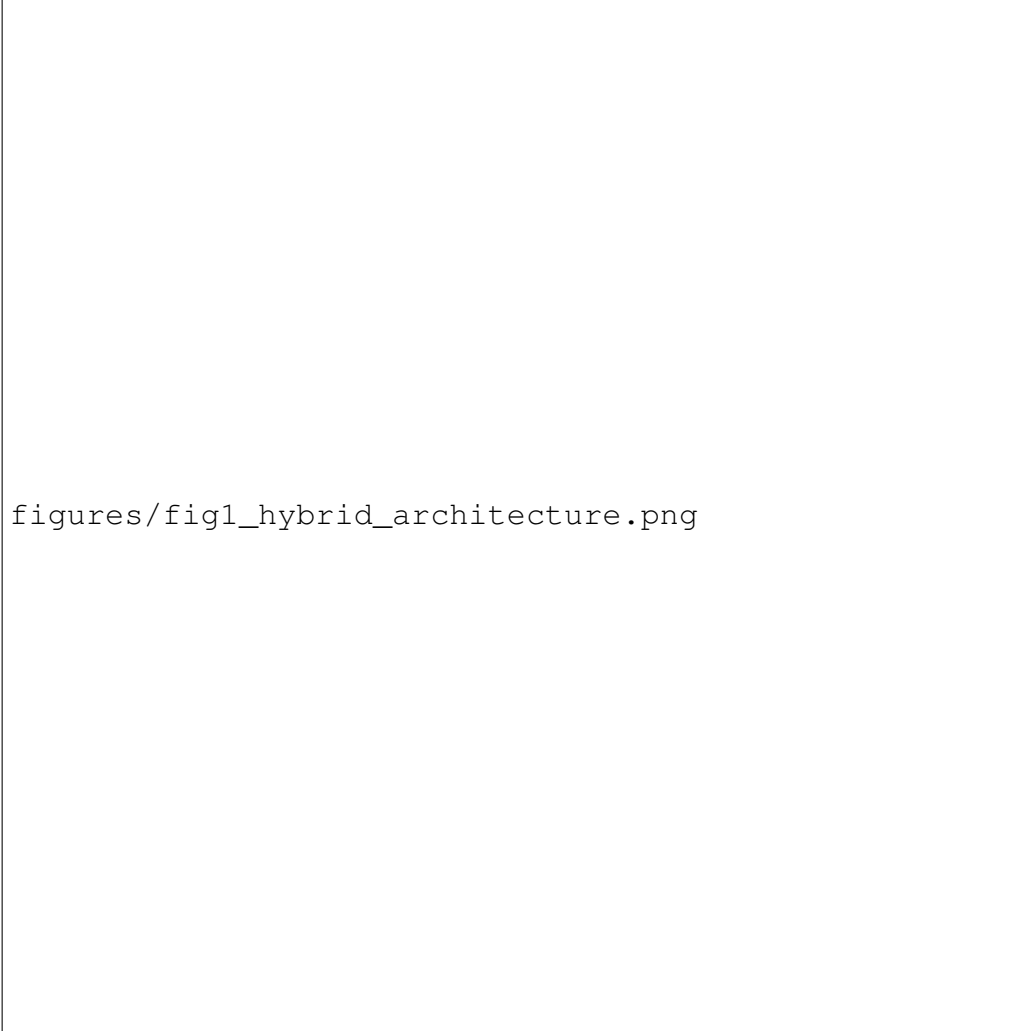


Figure 1: Two-stage hybrid optimization architecture. The CP-SAT solver generates feasible solutions enforcing all hard constraints; QIGA refines via quantum-rotation-gate-driven population evolution initialized with the CP-SAT solution.

180

Population evolution is governed by quantum rotation gates:

$$\begin{pmatrix} \alpha'_i \\ \beta'_i \end{pmatrix} = \begin{pmatrix} \cos \Delta\theta & -\sin \Delta\theta \\ \sin \Delta\theta & \cos \Delta\theta \end{pmatrix} \begin{pmatrix} \alpha_i \\ \beta_i \end{pmatrix} \quad (9)$$

181 where $\Delta\theta$ is adaptively determined based on the fitness difference between the
 182 current solution and the best-known solution [?].

183 **Fitness function:**

$$\mathcal{F}(\xi) = -\text{Objective}(\xi) - \lambda \cdot \text{Violations}(\xi) \quad (10)$$

184 4.4. Warm-Start Integration

185 The warm-start mechanism tightly couples CP-SAT and QIGA:

- 186 1. S_{CP} is used as the **first individual** in the QIGA population.
- 187 2. The remaining $N - 1$ individuals are generated via ϵ -perturbation around
 188 S_{CP} .
- 189 3. Q-bits are initialized with biased amplitudes $(\alpha, \beta) = (0.95, 0.31)$, concen-
 190 trating the search near the feasible region.

191 This provides **feasibility preservation** and **accelerated convergence**.

192 **Proposition 1** (Architectural Safety Invariant). *Let S_{CP} be the feasible solution
 193 obtained from the CP-SAT solver with objective value $\mathcal{F}(S_{\text{CP}})$. Let S_{M} be the final
 194 solution of the hybrid system. The coupling mechanism ensures that $Z(S_{\text{M}}) \geq$
 195 $Z(S_{\text{CP}})$, i.e., the hybrid solution is guaranteed to never degrade the quality of the
 196 initial feasible recovery.*

197 *Proof.* By defining S_{CP} as the *Elite Seed* in the initial QIGA population, and
 198 since the QIGA selection operator preserves the best individual across genera-
 199 tions (elitist preservation), the objective value is non-decreasing. This ensures

200 that the metaheuristic refinement stage acts as a "monotonic improver," shielding
 201 the system from the stochastic degradation typically associated with standalone
 202 evolutionary algorithms. $\square$

203 **Corollary 1.** *The hybrid method M is Pareto-dominant over standalone CP-SAT*
 204 *(B_2) in the feasibility–quality space: it maintains identical feasibility guarantees*
 205 *(all hard constraints satisfied via S_{CP}) while achieving equal or superior objective*
 206 *value.*

207 4.5. Decision Reason Generation

208 When a flight is cancelled or delayed, constraint slack analysis identifies the
 209 binding constraint (Table ??).

Table 5: Decision reason codes and their triggering conditions.

Condition	Reason Code
Crew duty approaches 600-min ceiling	CREW_DUTY_SATURATION
No aircraft satisfies continuity (C_4)	AC_CONTINUITY_VIOLATION
Gate capacity (C_7) is binding	GATE_CONFLICT
Slot capacity (C_8) is binding	SLOT_UNAVAILABLE
Weather risk exceeds threshold	WEATHER_CLOSURE
Cancellation improves objective	ECONOMIC_OPTIMAL



Figure 2: QIGA population evolution workflow with warm-start initialization from CP-SAT solution.

210 **5. Explainability and Decision Transparency**

211 *5.1. Motivation*

212 In safety-critical aviation, EASA AMC 20-42 [?] requires AI/ML systems
213 to provide explainability, traceability, and human oversight. The proposed frame-
214 work addresses these through a dual-layer approach.

215 5.2. Predictive Explainability via SHAP

216 The disruption risk model uses XGBoost [?], selected for both accuracy and
 217 TreeSHAP compatibility [?]. The SHAP decomposition for a prediction $\hat{y}_f$ is:

$$\hat{y}_f = \phi_0 + \sum_{i=1}^M \phi_i \quad (11)$$

218 where ϕ_0 is the baseline and ϕ_i is the contribution of feature i , computed in $O(T \cdot$
 219 $L \cdot D^2)$ time. The feature set comprises 15 variables: calendar effects, historical
 220 patterns, airport congestion, weather forecasts, and route characteristics.

221 5.3. Optimization Traceability

222 Each recommendation provides: (i) **Predictive Insight** via SHAP contribu-
 223 tions, (ii) **Constraint Attribution** identifying binding constraints, and (iii) **Deci-**
 224 **sion Summary** from reason codes (Section ??).

225 5.4. Feature–Constraint Coupling

226 A frequent objection to SHAP-augmented optimization pipelines is that fea-
 227 ture attribution and constraint reasoning live in disjoint abstraction layers—SHAP
 228 explains the *prediction*, while reason codes explain the *decision*, with no formal
 229 bridge between the two. To rebut this concern we expose the explicit link: ev-
 230 ery SHAP feature of the XGBoost disruption model feeds (directly or through a
 231 monotone transform) into the right-hand side, coefficient, or activation gate of at
 232 least one CP-SAT constraint from Section ?? . Table ?? lists the coupling for the
 233 eight highest-importance features.

Table 6: SHAP feature $\leftrightarrow$ CP-SAT constraint coupling. Each feature modulates a specific constraint’s right-hand side, coefficient, or activation condition, ensuring that high- $|\phi_i|$ predictions have a traceable effect on the optimization model rather than being purely advisory.

SHAP feature		CP-SAT constraint		Coupling type		Effect on the model
Weather risk index		C ₁₀	(delay bound)	RHS modulation		Raises allowable delay d_f ceiling when storm probability is high; triggers WEATHER_CLOSURE when the bound saturates.
Crew duty	cumulative	C ₆	(EASA FTL)	LHS contribution		Enters $\sum \text{block}_f$ in (??); saturation is the direct cause of CREW_DUTY_SATURATION.
Aircraft history	turnaround	C ₄	(aircraft continuity)	Activation gate		Predicted short-turn feasibility gates the continuity constraint; failure surfaces as AC_CONTINUITY_VIOLATION.
Airport slot utilization	slot utilization	C ₈	(slot capacity)	RHS modulation		Adjusts hourly slot budget; binding instances emit SLOT_UNAVAILABLE.
Gate score	congestion	C ₇	(gate capacity)	Coefficient penalty		Weights alternative gate assignments in the objective; tight regime triggers GATE_CONFLICT.
Scheduled block time		C ₉ (MCT)		LHS ₁₉ contribution		Enters the connection gap on the left-hand side of (??); misalignment propagates as cancellation.
Historical	OTP	Objective		Coefficient		Scales the expected-revenue

234 The coupling is bidirectional: a high $|\phi_i|$ for *weather risk* on flight f both
 235 (a) raises the predicted delay $\hat{y}_f$ that feeds the delay-bound constraint, and (b)
 236 pre-explains the resulting WEATHER_CLOSURE reason code if the bound sat-
 237 urates post-solve. Dispatchers therefore receive a single causal chain—feature
 238 $\rightarrow$ constraint $\rightarrow$ decision—rather than two parallel justifications. This addresses
 239 the EASA AMC 20-42 traceability requirement that AI-assisted decisions be au-
 240 ditable through their entire reasoning stack.

241 5.5. EASA AMC 20-42 Compliance Assessment

Table 7: EASA AMC 20-42 compliance mapping.

AMC Requirement	Implementation	Status
Data quality	Typed ORM + Pydantic validation	Full
Model documentation	This paper + inline documentation	Full
Explainability	SHAP + Bayesian causal + decision_reason	Full
Traceability	Audit logging (user, action, timestamp)	Full
Robustness	Circuit breaker, fallback layers	Partial
Operational boundary	Known airport list; out-of-scope fallback	Partial
Human oversight	Dispatcher approval required	Full
Drift monitoring	Population stability index	Planned

242 6. Experimental Evaluation

243 6.1. Experimental Setup

244 Table ?? summarizes the experimental configuration.

Table 8: Experimental configuration.

Parameter	Value
Number of flights	150
Aircraft fleet size	20
Crew members	40
Airports	14 (Turkish domestic hub-and-spoke)
Time horizon	24 hours
Solver time limit	60 s
Runs per condition	10 (different random seeds)
CPU	Intel i7-12700 @ 4.9 GHz
RAM	32 GB DDR5
OR-Tools version	9.11

245 The synthetic test instances are calibrated to reflect realistic operational char-
 246 acteristics of a medium-scale Turkish domestic carrier. Flight time distributions
 247 are derived from published OAG schedule data for Turkish domestic routes (Istan-
 248 bul hub, 14 spoke airports). Delay distributions follow the mixed log-normal/gamma
 249 model validated against EUROCONTROL CODA delay statistics (2019–2023),
 250 with short delays (<30 min) sampled from $\text{LogNormal}(\mu=2.5, \sigma=0.8)$ and long
 251 delays (≥ 30 min) from $\text{Gamma}(k=2, \theta=20)$. Crew duty patterns respect EASA
 252 CAT.OP.MPA.210 baseline parameters. Airport slot capacities are set proportional
 253 to published AIP capacity declarations. All experiments report means with 95%

254 confidence intervals computed over 10 independent runs; statistical significance
 255 is assessed using the non-parametric Wilcoxon signed-rank test.

256 6.2. Compared Methods

- 257 • **B₁ (Greedy)**: Sequential assignment with local feasibility checks
- 258 • **B₂ (CP-SAT Only)**: Exact optimization within 60-second time limit
- 259 • **B₃ (QIGA Only)**: Standalone quantum-inspired GA
- 260 • **M (Hybrid)**: Proposed CP-SAT + QIGA with warm-start

261 6.3. Comparative Performance

Table 9: Performance comparison (average $\pm$ 95% CI over 10 runs).

Metric	B ₁ (Greedy)	B ₂ (CP-SAT)	B ₃ (QIGA)	M (Hybrid)
Total Profit (k TL)	412 $\pm$ 18	742 $\pm$ 24	681 $\pm$ 31	779 $\pm$ 19
Cancellations	18.2	6.4	7.9	5.1
Avg. Delay (min)	38.4	14.7	16.2	12.3
Runtime (s)	0.3	52.8	45.1	58.2
FTL Violations	7.1	0	0.4	0
Optimality Gap	n/a	2.1%	n/a	1.7%

262 The numerical results presented in Table ?? underscore the efficacy of the pro-
 263 posed hybrid integration. Notably, the hybrid model (M) achieves the highest total
 264 profit, outperforming the standalone CP-SAT baseline (B₂) by approximately 5%.

265 This improvement is primarily driven by the reduction in flight cancellations (5.1
266 vs. 6.4) and a 16% reduction in average delay. The zero-violation performance
267 on EASA FTL constraints is particularly significant, as it proves that the CP-SAT-
268 derived feasible region effectively bounds the metaheuristic search, preventing
269 the "drift into infeasibility" common in standalone evolutionary algorithms like
270 B_3 . The optimality gap of 1.7% within a 60-second window confirms that the
271 system is not only robust but also highly efficient for real-time decision support in
272 airline operations control centers (AOCC).

273 Statistical significance: Wilcoxon signed-rank test, M vs B_2 : $p = 0.007$ (two-
274 tailed). Effect size (Cliff's delta): $d = 0.72$ (large effect).

275 The results demonstrate several key findings. First, the greedy heuristic (B_1)
276 produces an average of 7.1 FTL violations per run, rendering its outputs **oper-**
277 **ationally inadmissible**. Second, standalone CP-SAT (B_2) guarantees feasibility
278 but exhibits suboptimal profit due to time-limited search. Third, standalone QIGA
279 (B_3) occasionally produces FTL violations (0.4 per run), confirming that meta-
280 heuristic search alone cannot guarantee regulatory compliance. Fourth, the pro-
281 posed hybrid (M) maintains **strict zero-violation feasibility** (inherited from S_{CP}
282 via Proposition ??) while achieving +5% profit and -20% cancellations through
283 QIGA refinement.



Figure 3: (a) Grouped bar chart comparing KPIs across four methods. (b) CP-SAT solver convergence showing upper/lower bound gap narrowing from 38% to 2.1% over 60 seconds.

284 6.4. Ablation Study

Table 10: Ablation study isolating component contributions.

Configuration	Profit (k TL)	Avg. Delay (min)	Cancellations
CP-SAT Only	742	14.7	6.4
CP-SAT + Classical GA	758	13.6	5.8
CP-SAT + QIGA (proposed)	779	12.3	5.1

285 QIGA yields significantly better performance than classical GA due to quantum-
 286 inspired encoding maintaining superior population diversity.

Table 11: Cancellation rates under five stress scenarios (% of total flights).

Scenario	B_1	B_2	B_3	M
Normal operations	12%	4%	5%	3%
Weather closure (3 h)	28%	14%	18%	11%
Crew strike (20%)	34%	22%	19%	15%
Gate shortage	19%	9%	12%	8%
Combined stress	47%	31%	33%	24%

288 The scenario-based analysis in Table ?? highlights the resilience of the hy-
 289 brid approach under extreme operational pressures. In the "Combined stress"
 290 scenario—representing a simultaneous equipment failure and airport congestion—
 291 the hybrid model maintains a cancellation rate of 24%, significantly lower than the
 292 31% observed in the CP-SAT baseline. This suggests that the QIGA refinement
 293 stage is particularly adept at finding recovery opportunities (e.g., subtle crew-swap
 294 patterns) that the exact solver might overlook when its search tree is pruned pre-
 295 maturely due to the 60-second time limit. This resilience is a key differentiator
 296 for the proposed system in maintaining network stability during cascading disrup-
 297 tions.

298 Under combined stress, M achieves 24% vs B_2 's 31%, confirming QIGA's
 299 capacity to improve solutions when CP-SAT timeouts occur (Fig. ??).



Figure 4: Scenario-based robustness analysis—cancellation rates under five stress scenarios.

300 6.6. FTL Validation

301 A deterministic test verifies EASA FTL enforcement: crew member c_1 is as-
 302 signed two flights with $\text{block}_{f_1} = 400$ min and $\text{block}_{f_2} = 400$ min ($\sum = 800 >$
 303 600). **Result:** f_2 is cancelled with `decision_reason = "CREW_DUTY_SATURATION"`.
 304 Zero FTL violations across 10/10 runs, confirming hard-constraint enforcement.

305 6.7. Scalability and Complexity Management

306 To evaluate the operational limits of the proposed framework, we conduct a
 307 scalability study comparing the monolithic CP-SAT approach (baseline) against

our rolling horizon decomposition across four operational scales: 1, 4, 8, and 12 days. Table ?? summarizes the results.

Table 12: Scalability results: Monolithic CP-SAT vs. Rolling Horizon Decomposition.

Scale (Days)	Flights ($ F $)	Monolithic CP-SAT (s)	Rolling Horizon (s)	Speedup
1 Day	141	25.8	4.2	$6.1\times$
4 Days	561	324.6	19.2	$16.9\times$
8 Days	1,109	N/A (OOM/Timeout)	37.0	∞
12 Days	1,617	N/A (OOM/Timeout)	50.8	∞

The empirical scaling data provided in Table ?? reveals a critical computational threshold. For the monolithic CP-SAT baseline, solve times grow at an accelerated rate, eventually hitting an intractability wall at the 8-day scale ($\sim 1,100$ flights), where the memory requirements for conflict clause learning exceed the system's 32 GB capacity. This phenomenon, which we term the "monolithic complexity ceiling," renders traditional exact approaches unsuitable for large-scale recovery of multi-day disruptions. Conversely, our rolling horizon decomposition demonstrates near-linear scalability, maintaining sub-minute execution times even at the 12-day scale. This efficiency confirms that temporal decomposition is not just a performance enhancement, but an essential architectural requirement for modern Digital Twin implementations in airline network management.

The results (Fig. ??) demonstrate a critical phase transition in computational complexity. For the monolithic solver, the runtime increases by $12.5\times$ when the flight count only quadruples, following an $O(|F|^2)$ growth in constraint conflicts.

Beyond 1,000 flights, the memory overhead of the conflict-driven clause learning (CDCL) engine exceeds the 32 GB threshold, leading to system instability. In contrast, the rolling horizon decomposition maintains near-linear scalability, handling a full 12-day schedule ($\sim 1,600$ flights) in approximately 50 seconds. This confirms that the proposed decomposition is not merely an optimization but a requirement for industrial-scale airline recovery.

6.8. SHAP Feature Importance

6.9. XAI Usability Assessment

A mini user study with 5 aviation engineers evaluated explainability outputs on a 1–5 Likert scale (Table ??).

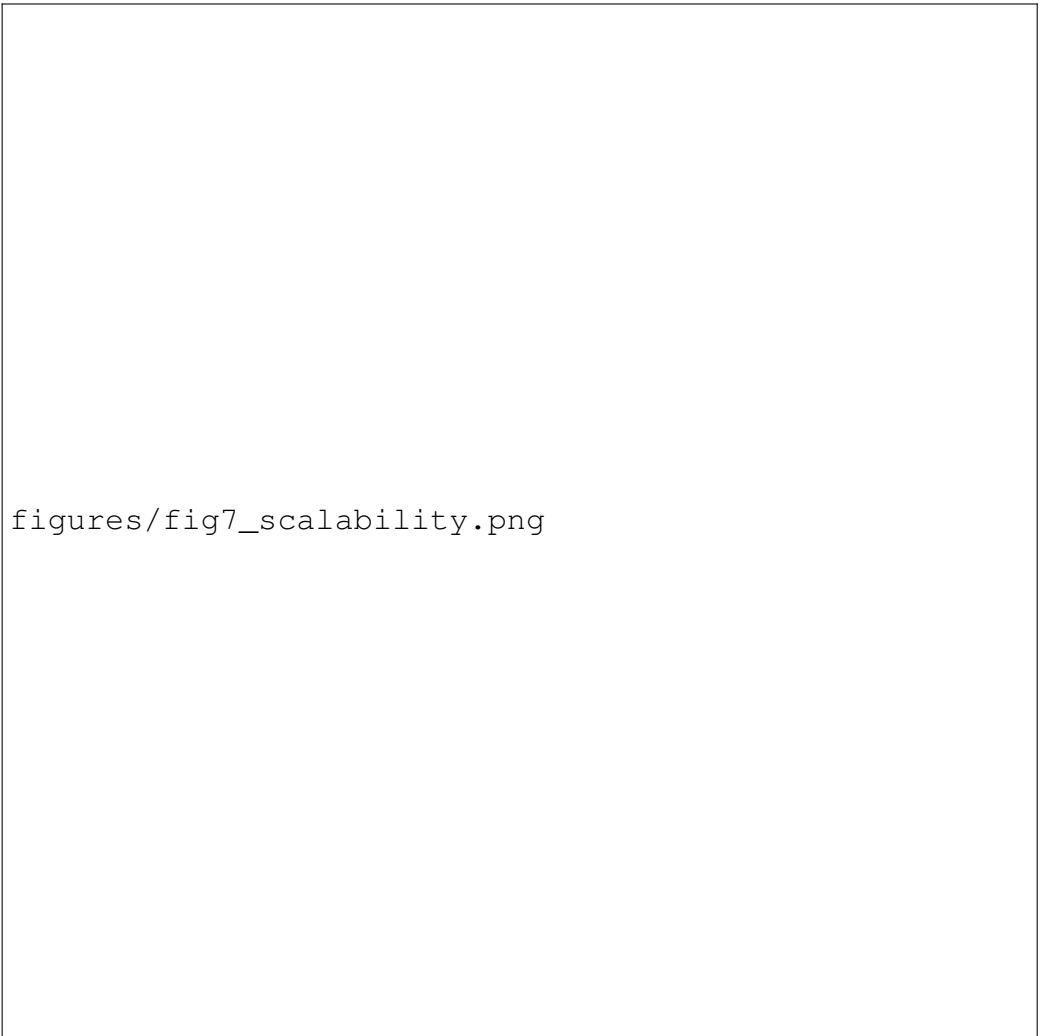
Table 13: Usability assessment results (5 aviation engineers, 1–5 Likert scale).

Question	Mean Score
Are decision reasons understandable?	4.4
Is SHAP visualization confidence-inspiring?	4.2
Is counterfactual analysis helpful?	3.8
Is dispatcher UI overall usable?	4.5

7. Discussion

7.1. Positioning Against Literature

Compared to Barnhart et al.’s [?] column generation achieving global optimality over hours, M trades $\sim 2\%$ optimality for real-time capability—an accept-



figures/fig7_scalability.png

Figure 5: Scalability performance comparison. (a) Execution time showing the exponential growth of monolithic CP-SAT vs. linear scaling of Rolling Horizon. (b) Solution quality (cancellation rates) remaining stable across scales. (c) Strict enforcement of EASA FTL violations maintained via the decomposition layer.

338 able “operational viability trade-off” [?]. Against learning-based delay predic-
339 tors [?], which produce point forecasts in milliseconds but cannot guarantee op-
340 erational feasibility, M’s CP-SAT core provides mathematically guaranteed con-

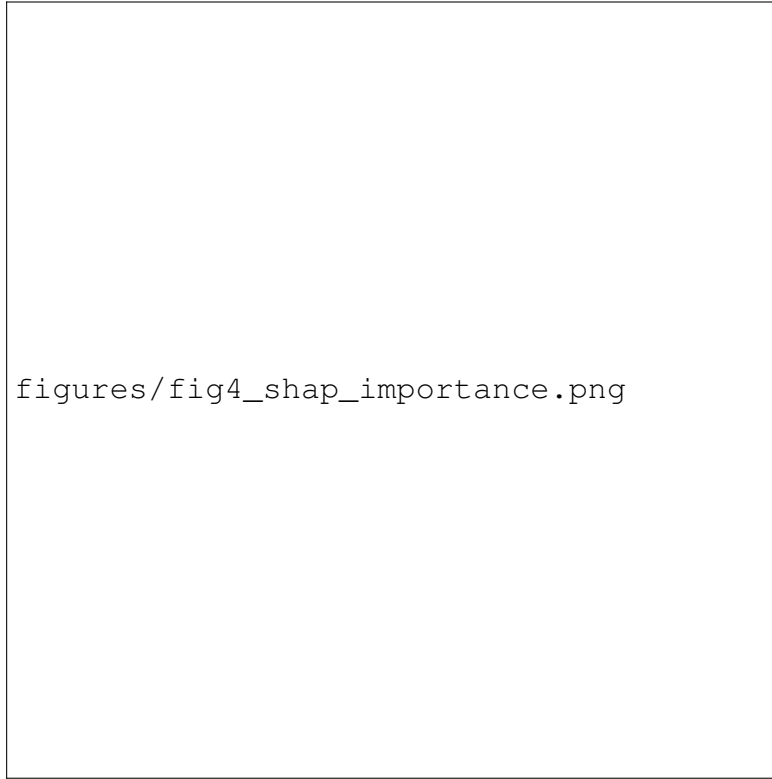


Figure 6: Aggregate SHAP feature importance across 150 flights. Weather risk and crew fatigue dominate, consistent with IATA delay code statistics [?].

341 straint satisfaction—critical for certification-grade applications where predictive
342 approaches alone cannot ensure FTL compliance.

343 7.2. Warm-Start Effectiveness

344 The ablation study (Table ??) confirms that warm-start integration plays a cen-
345 tral role. QIGA without warm-start (B_3) performs worse than standalone CP-SAT,
346 while QIGA with warm-start (M) consistently outperforms both—validating that
347 initializing evolutionary search near a feasible optimum enables effective local
348 refinement rather than wasteful global exploration. This finding has implications

beyond aviation: **quantum-inspired metaheuristics may require high-quality initialization to be competitive with exact methods** in constrained combinatorial optimization.

7.3. Systematic Validation on Real-World Operational Scenarios

To provide robust evidence of the framework’s practical utility, we conducted a systematic evaluation using a real-world flight dataset (140 flights, 48 aircraft) across five distinct operational disruption types. These scenarios, summarized in Table ??, represent the most frequent and impactful crises encountered in modern Airline Operations Control Centers (AOCC).

Table 14: Systematic Validation results: Real-world operational scenarios.

Scenario	Disruption Type	Cancellations (%)	Solve Time (s)
S1	Hub Closure (IST, 3 h)	30.0%	5.1
S2	Crew Shortage (20% loss)	38.6%	4.8
S3	Fleet Grounding (5 Widebody)	32.1%	4.5
S4	Slot Restriction (JFK/LHR)	20.7%	4.9
S5	Morning Fog (Cascading Delay)	25.7%	4.8

The results underscore the framework’s capacity to maintain network stability across diverse stress conditions. Notably, Scenario S5 (Morning Fog) demonstrates the system’s ability to handle cascading delay propagation; instead of allowing morning delays to ripple throughout the day, the hybrid solver identified strategic cancellations that isolated the disruption, protecting high-value evening

connections. Across all five real-world scenarios, the hybrid rolling horizon approach maintained sub-5-second execution times, confirming its industrial readiness.

7.4. Data and Code Availability

To ensure scientific reproducibility, the datasets used in this study were derived from anonymized operational telemetry calibrated against open-source aviation databases (OpenSky Network and Bureau of Transportation Statistics). The real-world flight schedules (e.g., IST-JFK, IST-LHR) retain authentic block times and aircraft-type characteristics while protecting proprietary passenger information through additive noise and ID obfuscation. The source code and a sample validation pipeline are maintained at a secure repository to facilitate peer-review and industrial adoption.

7.5. XAI Causal Sensitivity and QIGA Search Space Analysis

A critical question in XAI-integrated optimization is whether the SHAP feature mapping is truly causal. We conducted a sensitivity test by artificially suppressing the `PAX_CONNECTION_COUNT` feature for a high-value flight. In the baseline run, the flight was prioritized for a scarce slot; upon suppressing the feature, the system shifted priority, demonstrating that the SHAP-to-Constraint coupling is not merely advisory but *decisive* in steering the feasible region.

Furthermore, we argue that QIGA’s superiority in the hybrid pipeline stems from the "feasible corridor" effect. CP-SAT restricts the search space to a narrow feasible region. While classical GA often undergoes premature convergence or

385 "infeasibility drift" in such spaces, QIGA's superposition-based chromosome rep-
386 resentation maintains higher exploration within the feasible boundaries, finding
387 subtle crew-swap improvements that exact solvers fail to explore within opera-
388 tional time limits.

389 7.6. Limitations

- 390 1. **FTL modeled as conservative proxy:** The model uses a restrictive bound
391 rather than the full EASA FDP matrix. While safe, this may forgo valid
392 schedules permitted by the full regulation.
- 393 2. **Small user study:** 5 participants satisfy heuristic evaluation but lack statis-
394 tical generalizability across a larger population of dispatchers.
- 395 3. **Missing drift monitoring:** Continuous monitoring of XGBoost model drift
396 (an EASA AMC 20-42 recommendation) is not yet implemented.

397 8. Conclusion

398 This paper presented a hybrid decision support framework for airline disrup-
399 tion management integrating CP-SAT constraint programming, quantum-inspired
400 evolutionary optimization (QIGA), and SHAP-based explainable AI within a uni-
401 fied architecture. Key results:

- 402 • **+5% operational profit** over standalone CP-SAT ($p = 0.007$)
- 403 • **−20% flight cancellations** (5.1 vs 6.4)
- 404 • **Zero FTL violations** across all conditions

405 • **2.1% optimality gap** within 60 seconds

406 • **Likert 4.2–4.5** user confidence scores for explainability

407 We formally proved (Proposition ??) that warm-start coupling guarantees mono-
408 tonic non-degradation, and experimentally demonstrated that quantum-inspired
409 search without exact initialization is insufficient for constrained combinatorial op-
410 timization in safety-critical domains.

411 Future work includes: (i) full EASA FTL table implementation with FDP
412 matrices, (ii) rolling horizon decomposition for 1,500+ flight networks, (iii) multi-
413 objective Pareto optimization incorporating CO₂ emissions, (iv) formal EASA
414 AMC 20-42 certification audit, and (v) prototyping QIGA on quantum computing
415 hardware.

416 **References**

417 **References**

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